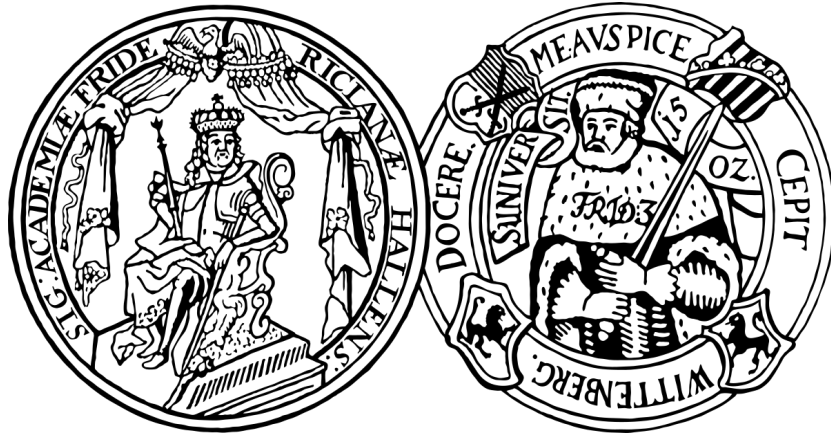


**Martin-Luther-University
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Masterseminar Seminar Paper

**Health Determinants of Panel Attrition:
A SOEP Logistic Regression Analysis**

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Abstract

Modelling data Attrition in the German SOEP, this seminar paper employs a logistic regression. First testing different model configurations using a frequentist approach, to then reestimate the best fitting model using a Bayesian framework. This model includes all sociodemographic controls as well as the 8 subscale health data in order to get more detailed results. Only mental health is found to have a significant impact on whether a person attrits from the survey. Looking closer at the effect mental health has on attrition given different interview types, high attrition modes correlate with a larger range of mental health's correlation on attrition. Leading to the results that while mental health is significant to predict attrition, it is even more so for interviews with people already being prone to attrit.

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1. Introduction

1.1 Background

Panel attrition refers to the loss of respondents from a longitudinal survey over time, when individuals who participated in earlier waves fail to respond in later waves. Attrition is a common feature of panel data and can arise for various reasons, including refusal to participate, inability to be contacted, illness, or death. As surveys progress over time, attrition typically accumulates, leading to a declining sample size in subsequent waves. In order to keep the sample varied, new groups of individuals need to be contacted in order to have a steady number of respondents.

From a methodological perspective, panel attrition is problematic because it can affect the representativeness of the remaining sample. If dropout occurs completely at random, attrition mainly reduces statistical power. However, if attrition is systematically related to observed or unobserved individual characteristics, estimates based on the remaining sample become biased. This issue is particularly relevant in socio-economic surveys, where attrition may correlate with income, employment status, demographic characteristics, or the focus of this paper, health.

In empirical research, panel attrition is commonly defined at the individual level as a transition from participation in one wave to non-participation in a subsequent wave. A respondent is typically classified as attrited if they permanently drop out of the survey (Siegers, Steinhauer, & Schütt, 2022).

1.2 Research Question

Longitudinal household surveys are a central data source in socio-economic research, as they allow the analysis of individuals and their situation over time. In every such survey exists the challenge of panel attrition, which occurs when respondents drop out of the study in later waves. Attrition not only reduces the sample over time, but leads to new people needing to be included in the survey, which is a costly process.

The German Socio-Economic Panel (SOEP) is one of the most widely used longitudinal surveys in Europe. Understanding the mechanisms driving attrition in the SOEP is therefore crucial for both applied research and survey design. Among potential determinants, health status plays a particularly important role, as poor health may reduce respondents' ability or willingness to continue participating in interviews.

This seminar paper analyses the impact of health factors on sample attrition in the SOEP. By modelling the probability of survey dropout as a function of health and socio-economic characteristics. The research question being:

To what extent are health factors related to the probability of sample attrition in the German Socio-Economic Panel?

The policy implications of the topic are particularly important. Cultivating a diverse sample in order to represent the population is a costly process, especially if new people have to be contacted to replace those leaving the survey. The process of gathering data is already costly, additionally when looking at the sample it is much cheaper to spend resources to keep existing participants rather than finding new ones (National Academies, 2013). With detailed results of people's tendencies to drop out of the survey, measures can be taken to preemptively accommodate the participant in order to retain them in the sample.

This paper will employ logistic regressions, estimating how health factors correlate with the outcome, whether a survey participant leaves the survey or not in the following year. In order to get a deeper understanding of marginal effects, Bayesian methodology will be utilised in order to sample model predictions. This provides estimates as well as credibility intervals used to create plots, which provide an intuitive and visual medium to understand how variables individually influence attrition. After answering the research question, the goal is to find tangible policy recommendations for the German SOEP based on the uncovered patterns.

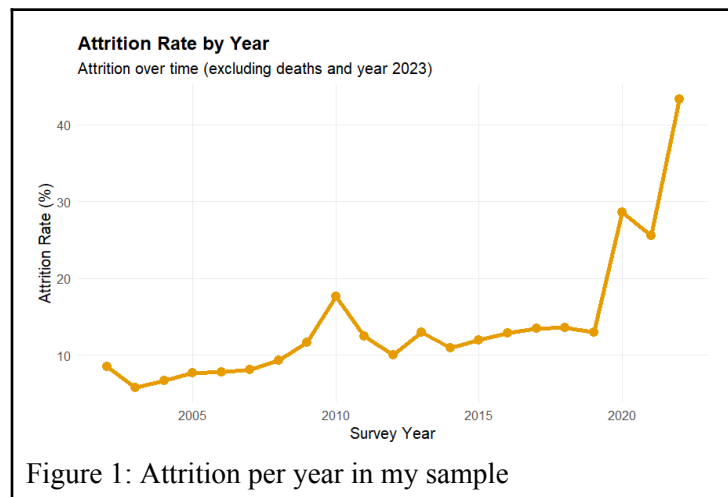
2. Method

2.1 Data and Sample

The concerned sample from the SOEP is limited from 2002 to 2023. The lower limit is set by the first occurrence of the health variables being measured, while 2023 is the last year observed in the dataset. Due to the way attrition is defined, the model uses data restricted to 2022, since it is not possible to measure permanent drop-out on the last possible observation.

For the given years, the sample comprises 427766 observations over 83407 unique individuals. This panel is not balanced since there are some missing values. Due to some variables being time consuming and costly to measure, such as grip strength, the sample is further reduced, with these being reported later for the relevant models.

Attrition over time in the sample can be observed in Figure 1. It ranges from around 8% before 2010, with it increasing to around 13% from 2009-2019, but it is especially high above 25% from 2020. More details can be seen in Figure 3. This increase will also be taken into account in the models.



2.2 Dependent Variable

In the German Socio-Economic Panel, the DIW defines attrition as the non-participation of an individual in wave $t+1$ after having participated in wave t , excluding cases where non-response is due to death. This definition focuses on survey-related dropout rather than unavoidable sample loss. Therefore the estimated effects other factors have on attrition are not influenced by death, leading to more usable results. The SOEP distinguishes between temporary non-response and permanent attrition (Siegers, Steinhauer, & Schütt, 2022). In this paper, attrition is defined as permanent attrition, with the exception of cases where the person answering has died.

2.3 Health Variables

In the SOEP, health-related quality of life is measured using the standardized Short-Form Health Survey(SF), more specifically the SF-12 which is composed of 12 questions related to different aspects of physical and mental wellbeing. The data appears in two ways, firstly through the Physical Component Summary (PCS) and the Mental Component Summary (MCS), as well as eight conceptual health categories, where the median is set at a score of 50 with a standard deviation of 10. PCS and MCS are constructed from the 12 items. Their derivation is based on factor analysis showing that responses to the SF-12 items are largely driven by two latent dimensions: physical and mental health (Grabka & SOEP Group, 2025).

The eight SF-12 health categories: physical functioning, role physical, bodily pain, general health, vitality, social functioning, role emotional, and mental health, are derived directly from individual questions from the SF-12. Each category is represented by one or two questions that capture a specific aspect of health, with weightings again being determined by factor analysis.

It must be noted that due to how the categories and the summary scores are constructed from the same underlying items, they are highly correlated. To avoid multicollinearity, analyses should therefore rely on either PCS/MCS or the set of category measures at one time, rather than combining them within the same specification.

Other available health metrics include Body Mass Index (BMI) and grip strength, which capture complementary aspects of physical health. BMI is included in standardized form to allow for comparability across individuals and to facilitate interpretation of effect sizes. BMI is measured on an absolute scale with limited interpretation per unit change, while standardization expresses deviations relative to the sample distribution.

Grip strength provides an objective measure of physical functioning and overall muscular strength and has been shown to be strongly associated with health outcomes. Lower grip strength is consistently linked with higher risk of mortality (Leong et al., 2015; Sun et al., 2024)), therefore it should be included alongside the other available health variables. This variable may indicate people's physical wellbeing, which is measured quantitatively instead of qualitatively

like the SF-12 survey. In the SOEP, grip strength is measured twice for each hand. This paper uses the maximum observed value across trials, due to it reflecting an individual's peak physical capacity, regardless of dominant hand or differences between measures. While some studies average multiple measurements, using the maximum is common in research when the goal is to capture true maximum performance as a predictor of health outcomes (Bohannon, 2004).

The health variables were measured on a biannual basis starting in 2002. As a result, the analysis is restricted to the period from 2002 to 2023, with missing observations addressed using a fill-forward approach to maintain continuity for the range of years.

2.4 Control Variables

Adding controls can be grouped more clearly as follows: demographic, socioeconomic, and survey-related factors. Demographic controls include age and gender, which account for life-cycle patterns and gender differences influencing attrition.

Migration background, which captures potential differences in survey participation for individuals born outside Germany. Jacobsen and Kühne (2021) show that those with a migration background have different attrition patterns due to moving more than average. Their research looks into introducing an app to get more response from these individuals, showing there is value in including this variable in the model, but also to look into the different modes of interview.

Socioeconomic and social stability controls include employment and partnership status, reflecting economic and social stability linked to continued participation, actual weekly working time, which complements employment status by indicating how busy participants are, and years of education or training, a proxy for long-term socioeconomic status.

Survey controls include interview mode fixed effects, which account for systematic differences related to the interview process, subsample fixed effects, which control for structural differences across SOEP samples, and year fixed effects, which absorb common time shocks and survey-wide changes influencing attrition across waves. Together, these controls help isolate the effect of health variables on attrition, accounting for well known predictors of attrition.

2.5 Statistical Models

The equation using the full extent of variables is described as:

$$\begin{aligned} \text{Logit}(P(\text{Attrition}_{it+1} = 1)) = & \\ & \beta_0 + \beta_1(PF_i) + \beta_2(RP_i) + \beta_3(BP_i) + \beta_4(GH_i) + \beta_5(VT_i) + \beta_6(SF_i) + \beta_7(RE_i) + \beta_8(MH_i) + \\ & \beta_9(BMI_std_i) + \beta_{10}(GripStr_i) + \beta_{11}(Age_i) + \beta_{12}(Female_i) + \\ & \beta_{13}(Employed_{it}) + \beta_{14}(MigBack_i) + \beta_{15}(Partner_{it}) + \beta_{16}(WorkH_{it}) + \beta_{17}(Education_i) + \\ & \Sigma\gamma_k(Mode_{it}) + \Sigma\delta_k(Sample_i) + \Sigma\theta_k(Year_t) + \epsilon_i \end{aligned}$$

Where:

i = individual, t = year

Health measures:

BMI_std, Grip strength, and eight SF-12 subscales (PF, RP, BP, GH, VT, SF, RE, MH)

Socioeconomic controls:

Age, Female, Employed, MigBack (migration background), Partner, WorkH (work hours per week), Education (years of education/training)

Fixed effects:

$\Sigma\gamma_k$ (Mode) with j = interview mode, $\Sigma\delta_k$ (Sample) with k = SOEP subsample, $\Sigma\theta_k$ (Year) with t = survey year.

Different models used to build understanding of which variables to include are specified as:

Model 1 removes grip strength, and all socioeconomic and fixed effects controls. It replaces the eight SF-12 subscales with two aggregated summary scores (MCS and PCS), retaining only basic demographic controls (age, gender) alongside BMI. Sample: N = 236,157.

Model 2 is identical to Model 1 but replaces the aggregated MCS and PCS scores with the disaggregated eight SF-12 subscales. All other variables remain the same. Sample: N = 236,157.

Model 3 adds grip strength to Model 1's specification (MCS, PCS, BMI, age, gender) but removes all socioeconomic controls and fixed effects. Due to missing grip strength data, the sample is substantially reduced. Sample: N = 55,726.

Model 5 is identical to Model 4 in sample size and control structure but replaces the eight SF-12 subscales with the aggregated MCS and PCS summary scores. Sample: N = 29,694.

3. Results

3.1 Frequentist Models

Five nested logistic regression models are estimated to examine health predictors of panel attrition. Models can only be meaningfully compared when they use identical samples, which limits direct comparisons to Models 1 versus 2 (N=236,157) and Models 4 versus 5 (N=29,694). All results referred to can be found in Table 1 (See Appendix).

Comparing Summary Scores to Subscales in the Full Sample (Models 1 vs. 2)

This initial comparison is done to gauge the difference between the 2 and 8 subscales of the health survey. Models 1 and 2 use identical samples and differ only in their measurement of health. Model 1 employs the two Physical and Mental Component summaries, while Model 2 disaggregates these into the eight subscales. Model 2 provides superior fit (AIC: 175,445 vs. 175,748), suggesting the subscales capture meaningful variation that the summary scores obscure.

Model 1 has both MCS and PCS being significant, while in Model 2, six of the eight subscales show significant effects on attrition. The conclusion to draw is that the 8 subscales do provide a better fit, but most importantly give more detailed results of which health factors predict sample attrition.

Adding Grip Strength (Model 3)

Model 3 introduces grip strength as an additional physical function measure but cannot be directly compared to Models 1-2 due to a 76% reduction in sample size (from 236,157 to

55,726). This reduction of sample is due to the lower amount of grip strength measurements, which is likely due to this measurement being more difficult and time consuming to take as opposed to the self reported questions of the SF-12 questions.

The reasoning to include model 3 is to see if the grip strength variable captures different information about physical fitness than the already existing variable. Grip strength proves significant ($\beta = -0.011$, $p < 0.001$) even controlling for MCS and PCS, suggesting it captures independent information about physical function not reflected in self-reported health measures. Both MCS and PCS remain significant in Model 3, as does BMI, though coefficients shift due to the different sample composition. While Model 3 demonstrates that grip strength adds predictive power, the sample selection limits generalizability of these findings.

Comparing Summary Scores to Subscales with Full Controls (Models 4 vs. 5)

Models 4 and 5 are directly comparable, sharing identical samples ($N=29,694$) and differing only in health measurement. Both include socioeconomic controls (employment, migration background, partner status, years in employment, years of education) and fixed effects for survey mode, sample type, and year. Model fit is almost identical (AIC: 17,492 vs. 17,490), indicating neither the subscale nor summary score approach offers a clear advantage once controls are added.

The key finding from this comparison is that most health effects disappear entirely with full controls. In Model 4, only the Mental Health subscale retains significance ($\beta = -0.009$, $p < 0.01$), while all seven other SF-12 dimensions become non-significant. In Model 5, both summary scores (MCS and PCS) are non-significant, with p-values of 0.144 and 0.411 respectively.

Both models show that grip strength and BMI, which were significant in simpler models, become non-significant with full controls. The mental health subscale in Model 4 is the sole exception, it detects a residual mental health effect that the MCS summary score in Model 5 misses. This suggests that while subscales generally offer no advantage over summary scores once controls are included, they may occasionally capture specific effects that aggregated measures obscure.

Although Model 5 has a marginally better AIC (17,490 vs. 17,492), Model 4 is selected as the primary specification because the subscale approach provides more detailed information than the summary score approach obscures. Model 4's advantage lies in its granularity: by disaggregating health into eight SF-12 subscales, it reveals that mental health specifically ($\beta = -0.009$, 95% CI: [-0.014, -0.003], $p = 0.003$) is the only health dimension retaining significance after comprehensive controls.

3.2 Full Bayesian Model

To assess the stability of Model 4's findings, the model is re-estimated using Bayesian logistic regression with weakly informative priors implemented through the `rstanarm` package. The Bayesian estimates converge almost perfectly with the frequentist results, confirming the findings are robust to the estimation method. For the mental health subscale, the Bayesian median coefficient is -0.009 with a 95% credible interval of [-0.015, -0.003], identical to the frequentist estimate of -0.009 with confidence interval [-0.014, -0.003] (See Table 2 in Appendix). All other health subscales similarly show negligible differences, all have Bayesian-frequentist differences under 0.001. BMI and grip strength remain non-significant in both approaches.

Bayesian diagnostics confirm model adequacy: all $\hat{R}$ values are below 1.02 indicating proper chain convergence, effective sample sizes exceed 400 for all parameters, and leave-one-out cross-validation shows all Pareto k values under 0.7, indicating good predictive performance without influential outliers.

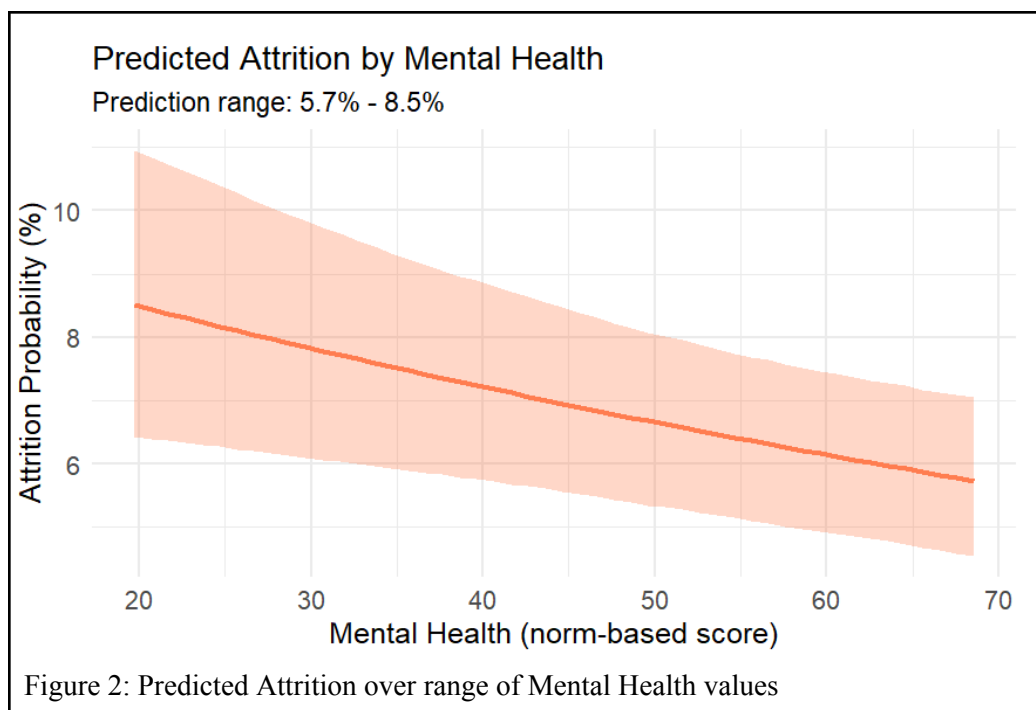
3.3 Interpreting Results

The downside of logistic regression is the interpretability of estimated coefficients. So far only coefficients and confidence / credibility intervals have been mentioned, with the purpose of comparing models, and to see whether results are different from 0. A more intuitive interpretation is in percentage change of odds, which can be found in Table 5 and 6 (See Appendix).

The most important results, whose credibility interval do not cover 0 can be interpreted as:

- Individuals with a migration background have 22% higher odds of attriting compared to those who do not.
- Each additional year of education reduces attrition odds by 5%.
- Each additional year of age reduces attrition odds by 1%.

Women show marginally lower attrition than men, with 10% reduced odds (95% CrI: [-21%, +2%]). However, the credible interval barely includes zero, suggesting this finding should be interpreted cautiously. The gender effect appears less robust than the socioeconomic predictors.



Mental health is the only health dimension significantly predicting attrition. One standard deviation improvement on the mental health subscale reduces attrition odds by 8%. By sampling the distributions of predicted values using the Bayesian model, and using median values for the other variables, Figure 2 shows the attrition probability over the range of mental health values.

At the lowest mental health score, people are expected to attrit at a rate of 8.5% for any given year, whereas individuals who have comparatively better mental health are only predicted to attrit 5.7% of the time.

3.4 Robustness Check

Attrition Right Censoring

Due to the construction of the attrition variable, there is a risk for it to be affected by right censoring. It may be the case that the last participant's last observed year was 2022, which would be considered as attrition. However, due to right censoring, with years later than 2023 not being observed, this attrition may only be temporary and should not be counted as such. This issue is

exacerbated by the fact that these last observed years are during the COVID-19 pandemic, which likely impacts the rate of SOEP answers. This can be seen in Figure 3, where 2020 - 2022 has a much higher rate of attrition. As a robustness check of the results, the model is reestimated, limiting observations to 2020, to see if the covid years have changed the estimated results.

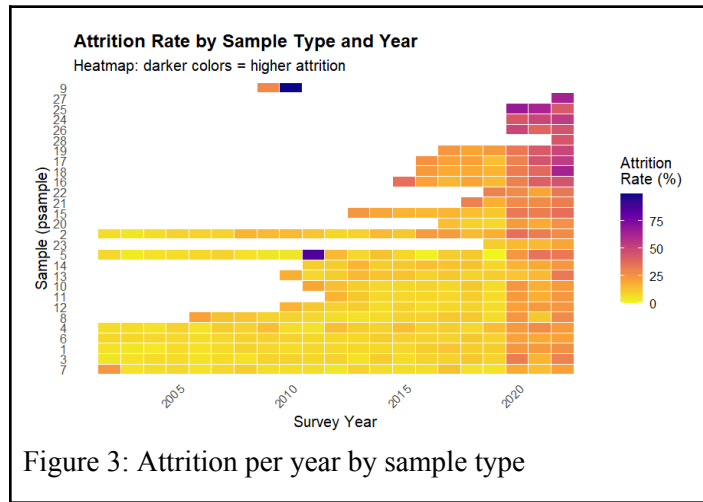


Figure 3: Attrition per year by sample type

Between the 2 models, the core findings are unchanged (See Table 3 in Appendix): mental health is the only significant health predictor in both specifications, with nearly identical effects. The robust model yields $\beta = -0.010$ (95% CrI: [-0.016, -0.003]) versus $\beta = -0.009$ (95% CrI: [-0.015, -0.003]) in the full model. All other health subscales remain non-significant. This stability confirms that the mental health-attrition link is not changed by the pandemic or recent years.

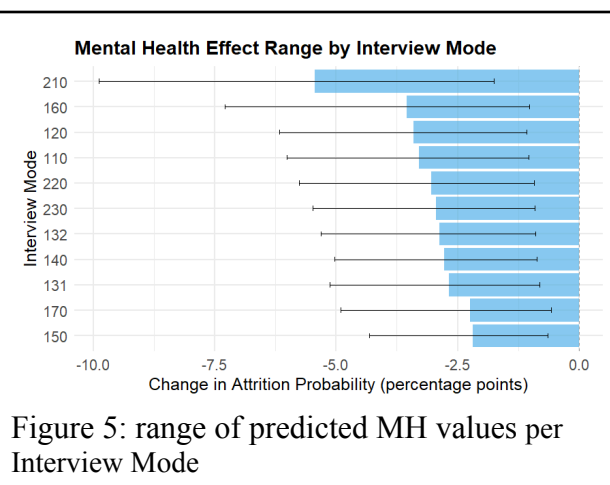
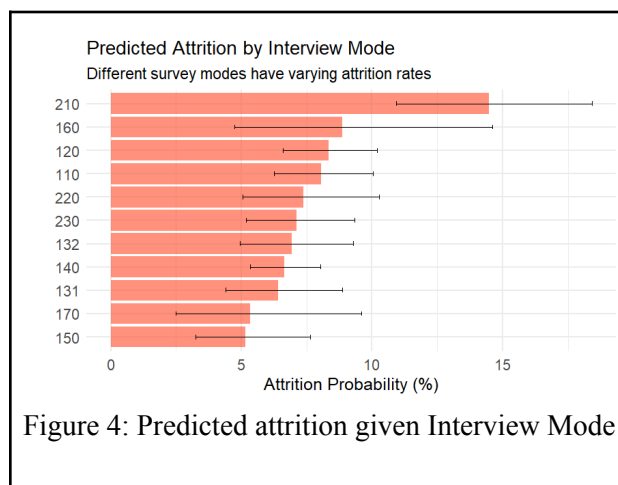
While health effects remain stable, some socioeconomic predictors change substantially when pandemic years are included. Migration background shows the largest shift: $\beta = 0.287$ (95% CrI: [0.187, 0.391]) pre-pandemic versus $\beta = 0.199$ (95% CrI: [0.115, 0.286]) in the full period. The weakening suggests pandemic disruptions affected all demographic groups more equally, temporarily reducing the elevated attrition typically observed among migrants. This could reflect universal challenges with survey participation during lockdowns. Women's protective effect similarly weakens from -0.144 to -0.108, suggesting gender gaps in attrition narrowed during the pandemic.

The main health factor predicting attrition holds across both models, with mental health being the sole health predictor of attrition after controls. This relationship is temporally stable despite right censoring affecting the attrition variable, and the pandemic-era changes which may have impacted socioeconomic attrition patterns.

4. Discussion

4.1 Further Exploration into Mental Health

Figure 4 shows the predicted attrition per interview type using the Bayesian model. The error bars are wide, however interview mode 210 has the largest attrition in the sample. This mode refers to surveys received home by people and sent back by post, with no interviewer present. Looking at Figure 5, where the highest and lowest extent of attrition prediction over the range of mental health values are represented according to the same interview modes. Note that despite also having large error bands representing the 95% credibility interval, there seems to be a correlation between modes with highest attrition, and the range of how much Mental health predicts attrition.

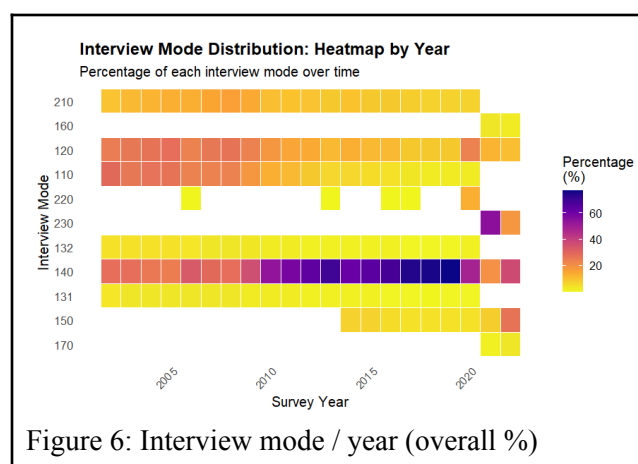


In Table 4 (See Appendix) are details of the different interview modes. My hypothesis is that the reason there is higher attrition is due to these modes having less “accountability”. Mode 210 is completed entirely without supervision from the comfort of one’s home which is why mental health would have a larger range of predicted effect. Similarly for the next on the list, mode 160

is done by phone, where it is easy not to pick up, and mode 120 is similar to 210, except the survey is hand delivered and retrieved. While other modes of interview have more “accountability”, for example mode 140 being a face-to-face interview, where respondents may feel more of an obligation to answer the survey and be relied upon.

Note that modes with less attrition are more likely to have self selection, for example, participants to undertake an interview at a particular location have chosen this option. Further research into interview mode may give better guidelines for survey designers to limit attrition.

Figure 6 shows which interviews are used every year, as well as the overall percentage this mode represents in that year. It may be difficult to draw a conclusion that the attrition of mode 210 is the reason it was discontinued, since corona times seems to have changed which modes are used. Nevertheless, there is more value in keeping modes where attrition is lower, which are also less correlated with changes in mental health.



4.2 Limitations of Results

The main limitation in the ability to detect health-attrition relationships is fundamentally limited by measurement timing. The SF-12 is administered only biennially, meaning health status in year t is used to predict attrition that could occur anytime in years $t+1$ or $t+2$. This temporal mismatch weakens estimated correlations and may specifically explain why physical health shows no effect. Acute health shocks like severe injuries or rapid functional decline could trigger dropout within weeks or months, well before the next scheduled health assessment. This paper's data structure implicitly assumes health changes gradually enough to be captured biennially, an assumption likely violated for sudden physical health events. Mental health, which may exhibit more stable trajectories or gradual deterioration, could be less affected by this measurement limitation, potentially explaining why only mental health remains significant in the models.

As mentioned previously, there is the issue of self-selection. In the case of someone suffering long-term from physical health limitations, they may be self-selecting into interview methods which are more convenient, such as telephone or mail surveys rather than face-to-face interviews. This adaptive behavior would allow physically impaired individuals to remain in the panel despite their health challenges, thereby attenuating any observable relationship between physical health and permanent panel dropout. The models include survey mode fixed effects to control for such selection, but cannot distinguish between individuals with poor physical health who persist in the panel or shift to a less demanding response mode.

5. Conclusion

The results indicate that mental health is the only health factor derived from the SF-12 that predicts panel attrition in the German SOEP, accounting for demographic and socioeconomic background, as well as survey fixed effects. Each standard deviation improvement of mental health is linked to an 8% decrease in leaving the panel in the subsequent year. Using the eight health subscales unveils this conclusion which would not be found only using the mental and physical component summaries of health. While grip strength has its place alongside the SF-12 data, capturing physical ability and health, no indicator of a person's physical wellbeing is correlated with dropping out of the sample in the subsequent year.

The lack of physical health results may be caused by the biennial measurement schedule not capturing acute health events. Another explanation could be self-selection into less demanding survey modes among those with physical limitations. These measurement constraints suggest that the null findings for physical health should not be interpreted as evidence that physical health doesn't matter for attrition, but rather that the data structure cannot properly capture this relationship.

There seems to be a correlation between attrition per interview mode, and the predicted range of effect mental health has on these. While error bars are wide, it may be the case that modes where there is less accountability to participate would see people more easily not respond due to poor

mental health. Further research could be undertaken to test this hypothesis. Understanding these mechanisms could inform targeted retention strategies.

Beyond health, the socioeconomic predictors capture a large degree of variation in attrition when looking at models without and with them included. Having a migration background is estimated to have 22% higher odds of attriting. Education emerges as a particularly strong factor against attrition, with each additional year reducing attrition odds by 5%. Age also shows a protective effect, with each additional year reducing attrition odds by 1%. Women show marginally lower attrition than men with 10% reduced odds, though the credible interval barely excludes zero, suggesting this gender effect is less robust than the education and migration findings. Employment status, partner status, and years in employment, while included as controls, showed credible intervals that included zero in the final model, indicating that the education and migration effects capture something beyond simple socioeconomic status or life stability.

While these findings come from the German SOEP, similar patterns likely exist in other household panels. The SF-12 is used internationally, where the challenge remains the same, to maintain participation from individuals experiencing mental health difficulties. However, the specific magnitude of effects may vary by cultural context, healthcare systems, and survey infrastructure. Such as interview mode. Replication in panels from diverse national contexts would establish whether mental health consistently predicts attrition or whether this is particular to the German setting.

This study demonstrates that mental health deserves special attention in both panel maintenance strategies and attrition analyses. The German SOEP should consider monitoring mental health and the other health variables more frequently than the current biennial schedule allows. Due to the nature of the SF-12 being answered similarly to other survey questions, unlike grip strength which needs a physical device to measure, there would be a low cost of making this part of the survey annual. Future attrition research should prioritize disaggregated health measures over summary scores and investigate whether measures can be taken to keep the answer rate high, while keeping the survey non-straining and easy for the respondents.

Appendix

Table 1: Frequentist Logistic Regression Models

Variable	Model 1	Model 2	Model 3	Model 4	Model 5
Physical Health					
PCS	-0.009*** (0.001)	—	-0.010*** (0.002)	—	-0.002 (0.003)
PF (Physical Functioning)	—	-0.012*** (0.001)	—	-0.000 (0.003)	—
RP (Role Physical)	—	-0.001 (0.001)	—	-0.004 (0.003)	—
BP (Bodily Pain)	—	-0.004*** (0.001)	—	-0.001 (0.003)	—
GH (General Health)	—	0.011*** (0.001)	—	0.002 (0.003)	—
Mental Health					
MCS	-0.008*** (0.001)	—	-0.012*** (0.001)	—	-0.003 (0.002)
VT (Vitality)	—	-0.004*** (0.001)	—	0.003 (0.003)	—
SF (Social Functioning)	—	-0.007*** (0.001)	—	0.004 (0.003)	—
RE (Role Emotional)	—	-0.002* (0.001)	—	0.000 (0.003)	—
MH (Mental Health)	—	0.001 (0.001)	—	-0.009** (0.003)	—
Other Health					
BMI (standardized)	-0.046*** (0.007)	-0.050*** (0.007)	-0.047** (0.015)	-0.013 (0.023)	-0.015 (0.022)
Grip Strength (max)	—	—	-0.011*** (0.002)	-0.002 (0.003)	-0.002 (0.003)
Demographics					

Age	-0.010*** (0.000)	-0.011*** (0.000)	-0.005*** (0.001)	-0.011*** (0.002)	-0.012*** (0.002)
Women	-0.093*** (0.013)	-0.099*** (0.013)	-0.259*** (0.041)	-0.108° (0.066)	-0.100 (0.066)
Socioeconomic					
Employed	—	—	—	-0.154 (0.350)	-0.150 (0.349)
Migration background	—	—	—	0.201*** (0.043)	0.202*** (0.043)
Partner	—	—	—	-0.030 (0.033)	-0.027 (0.033)
Years in employment	—	—	—	0.003° (0.002)	0.003° (0.002)
Years of education	—	—	—	-0.053*** (0.008)	-0.055*** (0.008)
Controls					
Survey mode FE	No	No	No	Yes	Yes
Sample FE	No	No	No	Yes	Yes
Year FE	No	No	No	Yes	Yes
Model Fit					
N	236,157	236,157	55,726	29,694	29,694
AIC	175,748	175,445	36,897	17,492	17,490

Note: Standard errors in parentheses. °p<0.10, *p<0.05, **p<0.01, ***p<0.001.

Table 2: Differences between the frequentist and Bayesian Estimations

Variable	Frequentist Estimate (SE)	Bayesian Median (MAD_SD)	Difference
Health Variables			
BMI (standardized)	-0.013 (0.023)	-0.012 (0.023)	0.001
Grip Strength (max)	-0.002 (0.003)	-0.002 (0.003)	0.000
PF (Physical Functioning)	-0.000 (0.003)	0.000 (0.003)	0.000
RP (Role Physical)	-0.004 (0.003)	-0.004 (0.004)	0.000
BP (Bodily Pain)	-0.001 (0.003)	-0.001 (0.003)	0.000
GH (General Health)	0.002 (0.003)	0.002 (0.003)	0.000
VT (Vitality)	0.003 (0.003)	0.003 (0.003)	0.000
SF (Social Functioning)	0.004 (0.003)	0.004 (0.003)	0.000
RE (Role Emotional)	0.000 (0.003)	0.000 (0.003)	0.000
MH (Mental Health)	-0.009** (0.003)	-0.009** (0.003)	0.000
Socioeconomic			
Age	-0.011*** (0.002)	-0.011*** (0.002)	0.000
Women	-0.108† (0.066)	-0.108† (0.064)	0.000
Employed	-0.154 (0.350)	-0.105 (0.353)	0.049
Migration background	0.201*** (0.043)	0.199*** (0.044)	-0.002
Partner	-0.030 (0.033)	-0.030 (0.033)	0.000
Years in employment	0.003† (0.002)	0.003† (0.002)	0.000
Years of education	-0.053*** (0.008)	-0.054*** (0.008)	-0.001
Intercept	-1.772*** (0.455)	-1.800*** (0.457)	-0.028

Table 3: Bayesian Model Comparison - Robust (2002-2019) vs. Standard (2002-2022)
(Median (MAD_SD) [95% CrI])

Variable	Robust 2002-2019	Standard 2002-2022	Difference
Health Variables			
BMI (standardized)	-0.045 (0.029) [-0.100, 0.011]	-0.012 (0.023) [-0.059, 0.033]	0.033
Grip Strength	-0.003 (0.003) [-0.010, 0.003]	-0.002 (0.003) [-0.007, 0.003]	0.001
PF (Physical Functioning)	0.001 (0.004) [-0.006, 0.009]	0.000 (0.003) [-0.007, 0.006]	-0.001
RP (Role Physical)	-0.004 (0.004) [-0.013, 0.004]	-0.004 (0.004) [-0.011, 0.003]	0.000
BP (Bodily Pain)	-0.003 (0.004) [-0.009, 0.004]	-0.001 (0.003) [-0.007, 0.005]	0.002
GH (General Health)	0.002 (0.004) [-0.005, 0.010]	0.002 (0.003) [-0.004, 0.008]	0.000
VT (Vitality)	0.003 (0.003) [-0.003, 0.010]	0.003 (0.003) [-0.002, 0.008]	0.000
SF (Social Functioning)	0.006 (0.004) [-0.002, 0.014]	0.004 (0.003) [-0.002, 0.010]	-0.002
RE (Role Emotional)	-0.001 (0.004) [-0.009, 0.007]	0.000 (0.003) [-0.006, 0.007]	0.001
MH (Mental Health)	-0.010 (0.003) [-0.016, -0.003]	-0.009 (0.003) [-0.015, -0.003]	0.001
Demographics			
Age	-0.011 (0.002) [-0.016, -0.007]	-0.011 (0.002) [-0.015, -0.007]	0.000
Women	-0.144 (0.080) [-0.300, 0.012]	-0.108 (0.064) [-0.239, 0.019]	0.036
Socioeconomic			
Employed	-0.258 (0.413) [-0.974, 0.684]	-0.105 (0.353) [-0.771, 0.649]	0.153
Migration background	0.287 (0.053) [0.187, 0.391]	0.199 (0.044) [0.115, 0.286]	-0.088
Partner	-0.049 (0.039)	-0.030 (0.033)	0.019
Years in employment	0.004 (0.002)	0.003 (0.002)	-0.001
Years of education	-0.042 (0.011)	-0.054 (0.008)	-0.012
Intercept	-1.827 (0.521) [-2.950, -0.842]	-1.800 (0.457) [-2.734, -0.895]	0.027
Model Diagnostics			
N (observations)	24,815	29,694	+4,879
Years included	2002-2019	2002-2022	+3

Table 4: Details of Different Interview Modes of the SOEP in the Sample

Mode	Method	Interviewer?	Location
210	Paper Mail	No	At home
160	Telephone	Yes	At home
120	Written/Paper	No	At home
110	Face-to-face	Yes	In-person
220	Telephone	Yes	At home
230	Telephone	Yes	At home
132	Oral + Written	Yes	In-person
140	Face-to-face	Yes	In-person
131	Written	Yes	In-person
150	Online/Web	No	At home
170	Computer	Yes	In-person

Table 5: Estimates transformed to Percentage change in Odds interpretation

Variable	% Change in Attrition	95% CI Lower	95% CI Upper
Migration Background	+22.2%	+12.2%	+33.1%
Female (vs Male)	-10.1%	-21.3%	+1.9%
Years of Education	-5.2%	-6.8%	-3.6%
Employed	-3.1%	-53.7%	+91.3%
Has Partner	-2.9%	-8.9%	+3.2%
BMI (1 SD increase)	-1.2%	-5.7%	+3.3%
Age (1 year)	-1.1%	-1.5%	-0.7%
Work Hours Per Week	+0.3%	-0.04%	+0.6%
Grip Strength (1 kg)	-0.2%	-0.7%	+0.3%

Table 6: Health estimates transformed to Percentage change in Odds interpretation

Variable	% Change (for 1 SD)	95% CI
Mental Health	-8.3%	[-13.5%, -2.9%]
Role Physical	-3.9%	[-10.1%, +2.6%]
Social Functioning	+3.9%	[-2.3%, +10.5%]
Vitality	+3.2%	[-2.2%, +8.6%]
General Health	+2.1%	[-4.2%, +8.5%]
Bodily Pain	-1.1%	[-6.9%, +4.8%]
Role Emotional	+0.4%	[-6.2%, +7.2%]
Physical Functioning	-0.05%	[-6.5%, +6.6%]

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Declaration of Authorship

I hereby declare that I have prepared the present final thesis paper independently, without unauthorized outside assistance, in compliance with the general principles of good scientific practice, and without using any sources and aids other than those specified. I have clearly labelled the texts, ideas, concepts, graphics, etc. taken directly or indirectly from external sources in my explanations as such and have provided complete references to the respective source. All other contents of this final thesis paper (text passages, illustrations, tables etc.) without corresponding references originate from me in the copyright sense. I assure that the final thesis paper has not yet been submitted and/or published in the same or a similar form as an examination in this or any other course of study. I am aware that the use of machine-generated content does not guarantee its quality and accuracy. I therefore affirm that I have only used generating AI tools as an aid and that my own intellectual and creative contribution predominates in this final thesis paper. I am fully responsible for the adoption of any machine-generated content I have used. I confirm that I have listed all AI-supported tools used and have explained and reflected on their use. The following AI-supported tools were used to create the final thesis paper:

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ChatGPT	In general	5	Checking if clarity is kept though the text, especially since parts are changed in different orders
	Bibliography	10	Finding papers: methodological research on Panel Sample and Attrition
Claude	Results	15	Help with coding, mainly plots for my results
	Appendix	15	Help to compile R results into the tables

Magdeburg, 06/02/26

Andan Grondt